

TOPIC CLASSIFICATION OF MEDICAL TRANSCRIPTIONS

A Comparison of Text-Only, NER-Enhanced,
and Negation-Aware Models

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PROJECT OVERVIEW

Problem Statement

- Can we automatically determine the medical topic of a dialogue/transcription using only text content?
- What distinctive terms do patients/doctors use in each topic?
 - 1,000 medical transcription samples
 - 5 specialties
 - Three classification approaches:
 1. Text Only (baseline)
 2. Text + NER Entities
 3. Text + NER (Removing Negated Entities)
 - Algorithms tested: Logistic Regression, SVM, Random Forest
 - Evaluation: Accuracy & F1-score



DATASET DESCRIPTION

- 5 specialties:
Cardiovascular/Pulmonary, Neurology, Radiology, Orthopedic, General Medicine
- Average transcript length: ~150–200 words
- Preprocessing steps:
 - Cleaning
 - Tokenization
 - Stopword removal
 - TF-IDF vectorization
- NER Models: ScispaCy en_core_sci_md + Negex
(negated entity detection)

APPROACH 1: TEXT-ONLY CLASSIFICATION

01

FEATURES USED:

- RAW TRANSCRIPTION TEXT
- PREPROCESSED + TF-IDF
- NO MEDICAL ENTITY EXTRACTION

02

MODELS TESTED:

- ✓ LOGISTIC REGRESSION
- ✓ SVM
- ✓ RANDOM FOREST



APPROACH 1: TEXT-ONLY CLASSIFICATION

Metric	Accuracy	F1 Score
Logistic Regression	0.630	0.622
SVM	0.610	0.610
Random Forest	0.545	0.541

Logistic Regression performed best.



APPROACH 2: ADDING NER ENTITIES

NER FROM SCISPACY:

- EXTRACTS MEDICAL CONCEPTS:
DISEASES, PROCEDURES, DRUGS,
ANATOMY
- ADDED AS EXTRA TOKENS:
- TEXT + ENTITY_1 + ENTITY_2 + ...

**ENTITIES = CLINICALLY
IMPORTANT INFORMATION**



APPROACH 2: ADDING NER ENTITIES

Metric	Accuracy	F1 Score
Logistic Regression + NER	0.630	0.625
SVM + NER	0.615	0.615
Random Forest + NER	0.555	0.554

Slight improvement over text-only.



APPROACH 3: REMOVING NEGATED ENTITIES

WHY REMOVE NEGATED?

- “NO FRACTURE”, “RULE OUT STROKE”, “NO EVIDENCE OF PNEUMONIA”
- THESE SHOULD NOT BE TREATED AS FEATURES INDICATING THE CONDITION.

METHOD:

- DETECT NEGATED ENTITIES
USING NEGEX
- REMOVE THEM BEFORE
VECTORIZATION



APPROACH 3: REMOVING NEGATED ENTITIES

Metric	Accuracy	F1 Score
Logistic Regression + NER (No Negated)	0.625	0.620
SVM + NER (No Negated)	0.620	0.620
Random Forest + NER (No Negated)	0.545	0.542

**Negation slightly improved SVM, but
small effect overall.**



SIDE-BY-SIDE COMPARISON

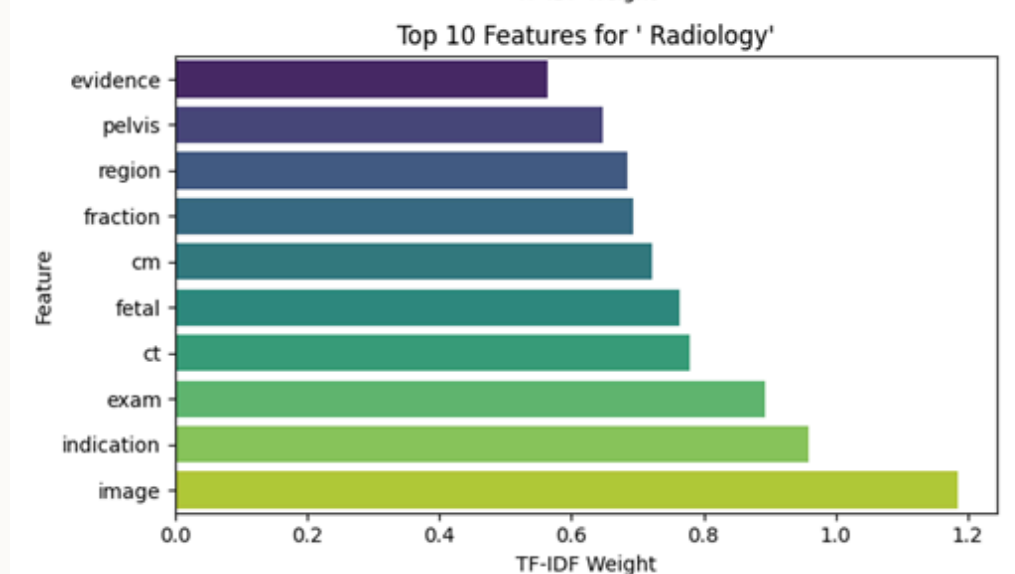
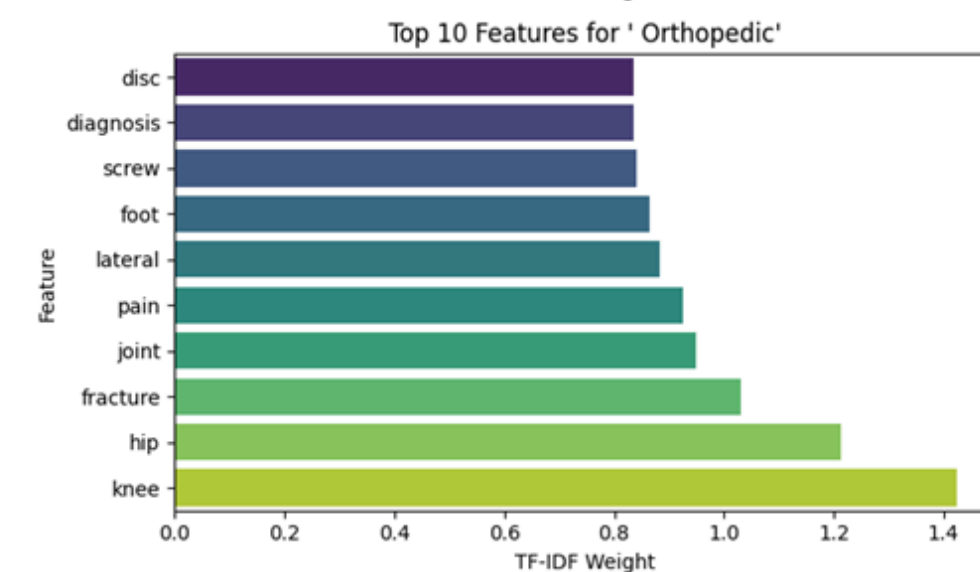
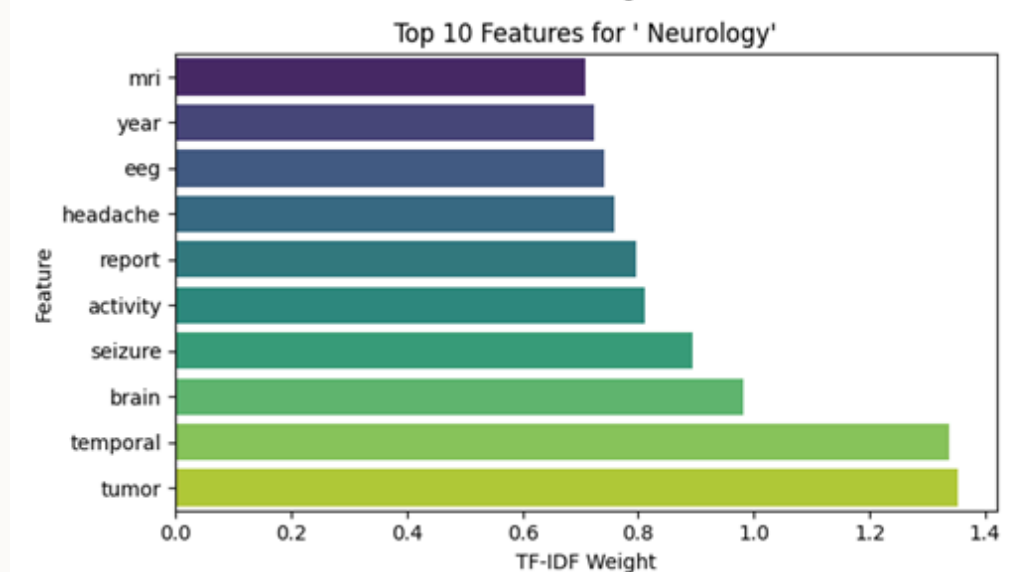
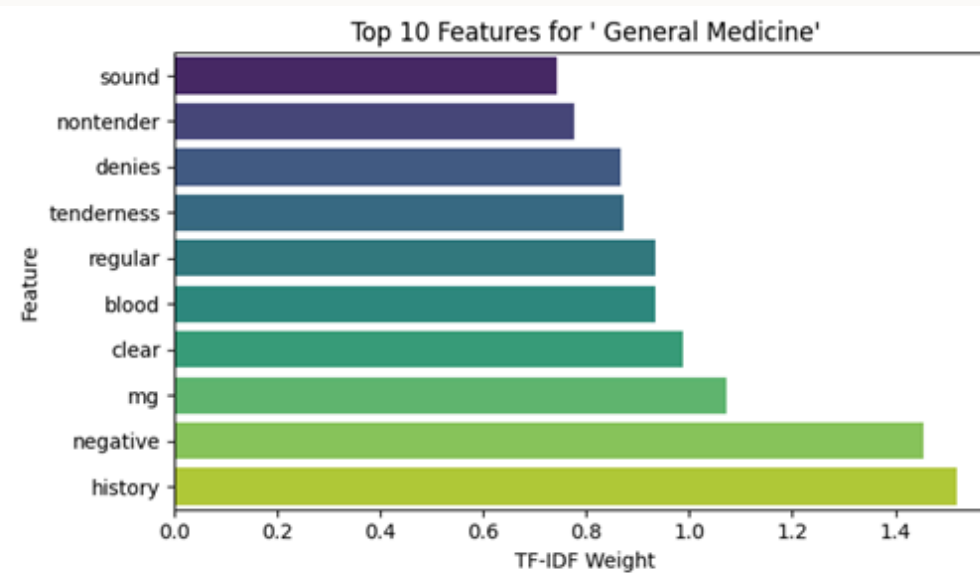
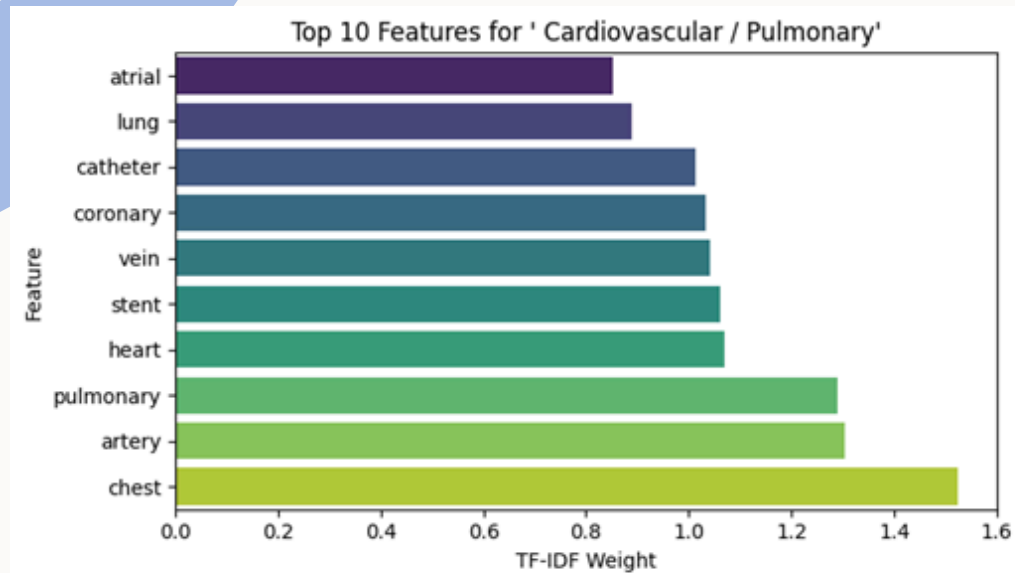
Model	Text Only F1	NER F1	NER w/o Negated F1
Logistic Regression	0.622	0.625	0.620
SVM	0.610	0.615	0.620
Random Forest	0.541	0.554	0.542

Main Conclusions

- NER improves performance a little
- Removing negated entities helps SVM
- Most gains are small → text alone already strong



TF-IDF FEATURE IMPORTANCE



- Cardiovascular / Pulmonary:
atrial, catheter, coronary, pulmonary, artery
- General Medicine:
denies, tenderness, mg, history
- Neurology:
headache, EEG, seizure, brain
- Orthopedic:
fracture, joint, hip, knee
- Radiology:
ct, image, pelvis, region

POSSIBLE IMPROVEMENTS

- Use transformers (BioBERT, ClinicalBERT)
- Add UMLS Concept Unique Identifiers (CUIs)
- Try hybrid representation: TF-IDF + embeddings
- Try deep-learning classifiers: CNN, LSTM
- Increase dataset size
- Apply SMOTE to balance classes
- Hyperparameter tuning on all models
- Use domain-adapted embeddings (scispaCy vectors)





What We Achieved

- Built three classification pipelines:
 - Text Only
 - Text + NER
 - Text + NER (Negated Entities Removed)
- Applied Logistic Regression, SVM, and Random Forest
- Implemented biomedical NER + clinical negation detection (NegEx)

Key Findings

- NER and negation-aware features enrich text representations, giving small but consistent improvements over raw text.
- Removing negated entities prevents misleading signals, especially in specialties with overlapping terminology.
- Best performing model:
- ★ SVM with Text + NER (No Negated) — most balanced F1 across all 5 specialties.



Conclusion

- Text remains the strongest base feature.
- Biomedical NER helps the model better understand clinical meaning.
- Negation handling adds clarity and reduces noise.
- Combined, they move the classifier closer to domain-aware intelligence.



Thank you